

Adapter Modules

Bottleneck FFN layers inserted into each Transformer block — base frozen, adapters trained

9 min

Duration

B3 – Apply

Bloom's

L3 – Advanced

Level

AI-FT-IN-TH-000009

ID

Adapter Modules — Definition

Definition (Houlsby et al., ICML 2019)

Adapter modules are small bottleneck feed-forward networks inserted into each Transformer block, with the base model weights completely frozen. Only the adapter parameters (~0.5–4% of total) are trained.

- Architecture: down-projection $W_{\text{down}} \in \mathbb{R}^{d \times k} \rightarrow$ non-linearity (GELU/ReLU) $\rightarrow$ up-projection $W_{\text{up}} \in \mathbb{R}^{k \times d}$, with a residual connection
- The residual initialises the adapter output to near-zero (W_{up} initialised to zero), preserving original model behaviour at training start
- Bottleneck dimension $k \ll d$: for $d=768$ (BERT-base), $k=64 \rightarrow 0.5\%$ of parameters
- Two insertion positions per Transformer block: after multi-head attention sublayer AND after feed-forward sublayer

Adapter Architecture — Transformer Block with Adapters

Transformer Block

Multi-Head Attention (FROZEN)

Adapter 1 $\leftarrow$ (TRAINED)

Layer Norm (FROZEN)

Feed-Forward Network (FROZEN)

Adapter 2 $\leftarrow$ (TRAINED)

Adapter Module Internals

Input $h \in \mathbb{R}^d$ (from frozen layer)

$W_{\text{down}} \in \mathbb{R}^{d \times k}$ (down-project to bottleneck k)

GELU / ReLU activation

$W_{\text{up}} \in \mathbb{R}^{k \times d}$ (up-project back to d)

+ Residual (skip connection from input h)

Output $h' = h + \text{adapter}(h)$

Adapter Variants — Extended Family

Variant	Key Innovation	Params	When to Use
Houlsby (original)	Adapters after attention + FFN; both positions	~3.6%	Multi-task BERT; highest quality baseline
Pfeiffer Adapter	Single adapter after FFN only; simpler	~0.8%	Lower overhead; minimal quality loss
AdapterFusion	Learnable composition of multiple task adapters	Multiple	Multi-task: combine N task adapters without re-training
Compacter	Hypercomplex multiplication for extreme compression	~0.05%	Very tight parameter budget
Llama-Adapter	Lightweight adapter for Llama-style instruction tuning	<1%	Instruction tuning with very few params

Adapter Modules vs LoRA — Practical Comparison

Adapter Modules

- Adds new layers to the architecture (forward pass overhead ~+5–15%)
- Fully separable: swap task adapter per forward pass (AdapterFusion)
- Standard insertion: two positions per layer
- Best for: multi-task settings, BERT-era models, server-side inference
- Native support: AdapterHub library (adapterhub.ml)

LoRA

- No new layers — modifies weight matrices (zero inference overhead after merge)
- Also separable: keep LoRA separate or merge into base
- Targets: specific weight matrices (Q, V, K, O, FFN)
- Best for: generative models (Llama, Mistral), on-device inference
- Native support: HuggingFace PEFT library

KEY REFERENCES

- Housby, N. et al. (2019). Parameter-Efficient Transfer Learning for NLP. ICML 2019.
- Pfeiffer, J. et al. (2020). AdapterFusion: Non-Destructive Task Composition for Transfer Learning. EACL 2021.
- Mahabadi, R.K. et al. (2021). Compacter: Efficient Low-Rank Hypercomplex Adapter Layers. NeurIPS 2021.



Next: Concept 11: PEFT Efficiency Trade-offs (AI-FT-IN-TH-000010)